جامعة الأميرة
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CS 111 - Lab

Programming Language 2
Computer Science Department
2026

Lab 7
Interfaces and Abstract Classes

Lab Objectives:

In this lab, a student will practice

• Reading part of a UML class diagram • Creating a class. • Declaring instance variables, constructor and accessors and mutators methods. • Implementing polymorphism	• Method overriding • Using ArrayList • **Abstract Classes** • **Interfaces** • Writing a test application to demonstrate the capabilities of another classes.

Lab Exercise – Food Ordering System

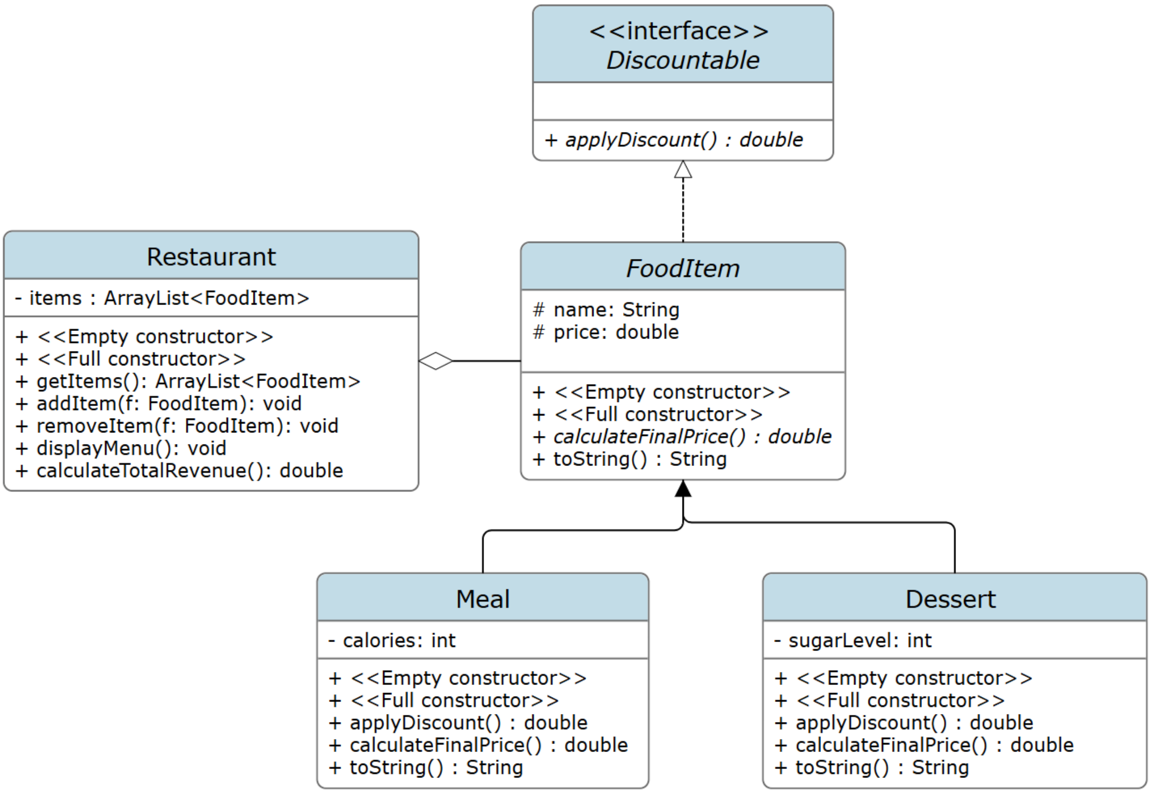
Given the following UML, implement all the classes in the same package with all needed instance variables, constructors (including empty constructors where applicable), accessor and mutator methods. Then write a test application to demonstrate the system capabilities.

The system represents a **Restaurant** that stores different types of food items.

Each item has a name, category, and price.

The restaurant uses an ArrayList<FoodItem> to store items.

Note: Add getters and setters where necessary.



In Discountable Interface

1. applyDiscount() :Returns the discount rate for the item

In FoodItem Class (Abstract)

1. calculateFinalPrice()→ Abstract method
2. toString()→ Returns food item information

In Meal Class

1. Override applyDiscount() → Returns **0.10**
2. Override calculateFinalPrice() → price after discount
3. Override toString() → Includes calories

In Dessert Class

1. Override applyDiscount() → Returns **0.20**
2. Override calculateFinalPrice()
   → price after discount
   → Add **+15** if sugarLevel > 50
3. Override toString() → Includes sugarLevel

In Restaurant Class

1. getItems() → Returns all items
2. addItem(FoodItem f) → Adds an item
3. removeItem(FoodItem f) → Removes an item
4. displayMenu() → Prints all items
5. calculateTotalRevenue() → Returns total final price of all items

In the Main Class

1. Create a Restaurant object
2. Create:
   - Two Meal objects
   - One Dessert object
3. Add items to the restaurant
4. Display all items
5. Print total revenue
6. Remove one item
7. Display menu again